

Section B

$$1.1) \quad g'(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}$$

$$\lim_{h \rightarrow 0} \frac{(x+h)(f(x+h)) - xf(x)}{h}$$

$$\lim_{h \rightarrow 0} \frac{xf(x+h) + hf(x+h) - xf(x)}{h}$$

$$\lim_{h \rightarrow 0} \left(\frac{xf(x+h) - xf(x)}{h} + \frac{hf(x+h)}{h} \right)$$

$$\lim_{h \rightarrow 0} \frac{xf(x+h) - xf(x)}{h} + \lim_{h \rightarrow 0} f(x+h)$$

$$= x \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + f(x)$$

$$= x f'(x) + f(x)$$

$$1.2) \quad f'(x) = e^{\sqrt{x}} \cdot \frac{1}{2} x^{-\frac{1}{2}} + 1 = \frac{e^{\sqrt{x}}}{2\sqrt{x}} + 1$$

to determine where the slope is 1, we let

$$f'(x) = \frac{e^{\sqrt{x}}}{2\sqrt{x}} + 1 = 1$$

$$\text{or } \frac{e^{\sqrt{x}}}{2\sqrt{x}} = 0 \quad \text{This is not true for any } x$$

$\Rightarrow$ the slope is never 1



$$1.3) \quad f(x) = \ln(4-x^2)$$

(Domain of $\ln$ is $(0, \infty)$)

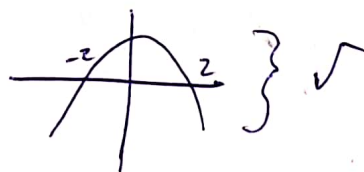
$f(x)$ is increasing where $f'(x) > 0$

$$f'(x) = \frac{-2x}{4-x^2} \quad \checkmark$$

$$f'(x) = \frac{-2x}{4-x^2} > 0 \Rightarrow -2x > 0 \Rightarrow x < 0 \quad \checkmark$$

since $4-x^2 > 0$

$$\Rightarrow x \in (-2, 2)$$



$$\Rightarrow x \in (-\infty, 0) \cap (-2, 2)$$

$$\Rightarrow x \in (-2, 0)$$

f is increasing on $(-2, 0)$ $\checkmark$

$$2.1) \quad \lim_{x \rightarrow 0} \frac{a^x - b^x}{c^x - d^x} \quad \left(\frac{0}{0}\right) \quad \checkmark$$

$$\lim_{x \rightarrow 0} \frac{a^x \ln a - b^x \ln b}{c^x \ln c - d^x \ln d} \quad \checkmark \quad \text{L'Hosp.} \quad \checkmark$$

$$= \frac{a^0 \ln a - b^0 \ln b}{c^0 \ln c - d^0 \ln d}$$

$$= \frac{\ln a - \ln b}{\ln c - \ln d} \quad \checkmark$$

$$2.2 \quad \lim_{x \rightarrow \infty} \frac{\sqrt{x^2+1}}{3x-5} \quad \left(\frac{\infty}{\infty}\right)$$

$$\lim_{x \rightarrow \infty} \frac{\sqrt{x^2(1 + \frac{1}{x^2})}}{x(3 - \frac{5}{x})}$$

$$\lim_{x \rightarrow \infty} \frac{\sqrt{x^2} \sqrt{1 + \frac{1}{x^2}}}{x(3 - \frac{5}{x})}$$

$$\lim_{x \rightarrow \infty} \frac{x \sqrt{1 + \frac{1}{x^2}}}{x(3 - \frac{5}{x})}$$

$$= \lim_{x \rightarrow \infty} \frac{\sqrt{1 + \frac{1}{x^2}}}{3 - \frac{5}{x}} = \frac{1}{3}$$

$$\sqrt{x^2} = \begin{cases} x & \text{if } x > 0 \\ -x & \text{if } x < 0 \end{cases}$$

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2+1}}{3x-5}$$

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2} \sqrt{1 + \frac{1}{x^2}}}{x(3 - \frac{5}{x})}$$

$$\lim_{x \rightarrow -\infty} \frac{-x \sqrt{1 + \frac{1}{x^2}}}{x(3 - \frac{5}{x})}$$

$$= \lim_{x \rightarrow -\infty} \frac{\sqrt{1 + \frac{1}{x^2}}}{3 - \frac{5}{x}}$$

$$= -\frac{1}{3}$$

$\Rightarrow y = \frac{1}{3}$ & $y = -\frac{1}{3}$ are horizontal asymptotes

Q3

3.1

$$\overrightarrow{OA} = \langle 6, 0, k \rangle$$

$$\overrightarrow{OB} = \langle 1, k, -3 \rangle$$

$$\overrightarrow{OA} \perp \overrightarrow{OB} = \overrightarrow{OA} \cdot \overrightarrow{OB} = 0$$

$$\Rightarrow \langle 6, 0, k \rangle \cdot \langle 1, k, -3 \rangle = 0$$

$$6 + 0 - 3k = 0$$

$$k = 2$$

$$\text{if } k=2, \overrightarrow{OA} \perp \overrightarrow{OB}$$

3.2

$$\overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OB}$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$$

$$= \langle 1, 1, -3 \rangle - \langle 6, 0, 2 \rangle$$

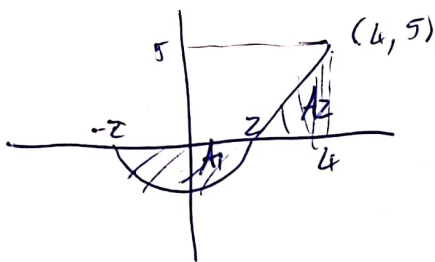
$$= \langle -5, 1, -5 \rangle$$

$$|\overrightarrow{AB}| = |\langle -5, 1, -5 \rangle|$$

$$= \sqrt{25 + 1 + 25}$$

$$= \sqrt{51}$$

Section C



Q.1) $f(x) = \begin{cases} -\sqrt{4-x^2} & \text{if } x \in [-2, 2] \quad \checkmark \\ \frac{5}{2}x - 5 & \text{if } x \in (2, 4] \quad \checkmark \end{cases}$

$$x^2 + y^2 = 4$$

$$y^2 = 4 - x^2$$

$$y = -\sqrt{4-x^2}$$

$$y = mx + c$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{5}{2}$$

$$y = \frac{5}{2}x + c$$

Q.2) Total Area = $A_1 + A_2$

$$= \frac{1}{2}\pi r^2 + \frac{1}{2}bh$$

$$= \frac{1}{2}\pi (2)^2 + \frac{1}{2}(2)(5)$$

$$= 2\pi + 5$$

Q.3) $\int_{-2}^4 f(x) dx$

$$= -A_1 + A_2$$

$$= -2\pi + 5$$

Q5

5.1 1. f is continuous on $\mathbb{R}$

f is contin. and ... on $[-2, 2]$ ✓

2. Critical numbers: $f'(x) = 4x^3 - 12x^2$

$$f'(x) = 0 = 4x^3 - 12x^2$$

$$4x^2(x-3) = 0$$

$$x = 0 \quad \text{or} \quad x = 3$$

but $x \notin [-2, 2]$

3. $f(-2) = 48$ $f(2) = -16$ ✓

$$f(0) = 0$$

4) Abs. min. of f on $[-2, 2]$ is $f(2) = -16$

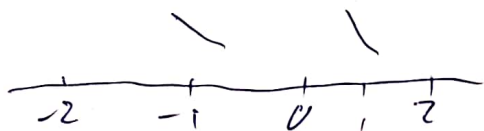
Abs. max is $f(-2) = 48$ ✓

5.2 Local max/min where $f'(x) = 0$

from (5.1) : we have a critical point at $x=0$

$$f'(-1) = < 0$$

$$f'(1) = < 0$$



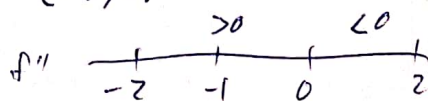
$\Rightarrow f$ is decreasing on $(-2, 0)$ and on $(0, 2)$

$\Rightarrow$ no local extremas on $(-2, 2)$

$$f''(x) = 12x^2 - 24x = 0$$

$$12x(x-2) = 0$$

$$x = 0 \quad \text{or} \quad x = 2$$



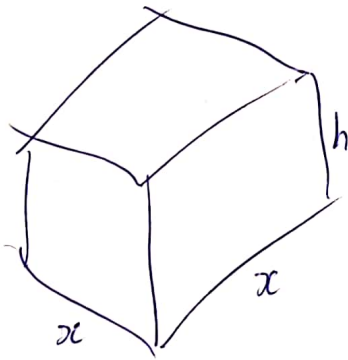
$$f''(-1) = > 0$$

$$f''(1) < 0$$

$\Rightarrow f$ changes concavity at $x=0$

$\Rightarrow f$ has inflection point at

$x=0$, inflection point: $(0, f(0)) = (0, 0)$



$$x^2 h = 32000$$

$$h = \frac{32000}{x}$$

$$\text{outside area} = l^2 + 4lh$$

$$= x^2 + 4x \left(\frac{32000}{x^2} \right)$$

$$= x^2 + \frac{4(32000)}{x}$$

We need A to be a min.

$$\Rightarrow A'(x) = 0$$

$$0 = 2x + 4(32000)(-x^{-2})$$

$$2x - \frac{4(32000)}{x^2} = 0$$

$$x = 40 \text{ cm}$$

$$h = 20 \text{ cm}$$

$\therefore$ Dimensions of the box, such that the outside area is a minimum is

$$x = 40 \text{ cm}, h = 20 \text{ cm}$$